Measurement Strategies in Quantum Computing

Why Measuring All Qubits is Optimal in Modern Algorithms

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Spring 2026

Motivation

Central question

Why is it preferable to measure all qubits rather than mapping observables onto a single qubit?

- Traditional textbooks emphasize single-qubit readout.
- Modern quantum algorithms (VQE, QML) rely on full-register measurement.

Quantum measurement structure

An n-qubit state is

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle.$$

Measurement in the computational basis yields:

$$p(x) = |c_x|^2.$$

Key observation

The outcome is a **joint probability distribution** over all qubits.

Theorem: Information loss in partial measurement

Theorem

Let ρ be an n-qubit state. Measuring only a subset of qubits yields marginal distributions which do not uniquely determine ρ or its multi-qubit correlations.

Proof.

Let ρ be entangled:

$$\rho \neq \rho_1 \otimes \rho_2.$$

Then in general:

$$\langle Z_i Z_j \rangle \neq \langle Z_i \rangle \langle Z_j \rangle.$$

Measuring only qubit i gives access to $\langle Z_i \rangle$, but not to joint correlators.
Hence the reduced statistics are insufficient to reconstruct expectation values of multi-qubit observables. $\square$

Corollary: Necessity of full measurement

Corollary

To estimate expectation values of general observables,

$$H = \sum_k c_k P_k,$$

with P_k multi-qubit operators, one must measure all qubits involved.

$$\langle P_k \rangle = \langle Z_{i_1} Z_{i_2} \cdots X_{j_1} \rangle$$

- Requires joint measurement outcomes
- Cannot be reconstructed from single-qubit marginals

Theorem: Circuit-depth optimality

Theorem

Let O be a multi-qubit observable. Any protocol that maps O onto a single qubit prior to measurement requires a unitary transformation whose circuit depth is at least proportional to the support size of O.

Sketch.

Mapping O to a single qubit requires a unitary U such that

$$U^\dagger Z_1 U = O.$$

If O acts on k qubits, then U must entangle these qubits with qubit 1.
Thus:

- Requires at least $\mathcal{O}(k)$ entangling gates
- Increases circuit depth and noise

Direct measurement avoids this overhead. □

Hardware argument

Asymmetry in cost

- Quantum gates: noisy and expensive
- Measurements: comparatively cheap and parallelizable

Thus:

$$\text{Adding gates (basis rotations, mappings)} \gg \text{measuring more qubits}$$

Deferred measurement principle

Principle

Measurements can be deferred until the end of a quantum circuit without changing outcome probabilities.

$$\Rightarrow \text{One can measure all qubits at the end}$$

This supports the modern paradigm of full-register measurement.

Variational algorithms

In hybrid algorithms:

$$\text{prepare state} \rightarrow \text{measure all qubits} \rightarrow \text{classical post-processing}$$

- Introduced with VQE (2014)
- Core workflow: sampling full bitstrings

Preskill (NISQ paradigm):

prepare an n-qubit state and measure all qubits, *see*

Why textbooks differ

- Early algorithms (Deutsch–Jozsa, phase estimation)
    - Encode answer in a single qubit/register
- Analytical simplifications:
    - Easier to track one observable
- Phase kickback techniques:
    - Transfer information to one qubit

Conclusion

These are **pedagogical or theoretical constructions**, not optimal experimental strategies.

Final theorem: Optimal measurement strategy

Theorem

For variational and sampling-based quantum algorithms, measuring all qubits in the computational basis minimizes circuit depth while preserving full information about the quantum state.

Argument.

- Full measurement yields complete probability distribution $p(x)$
- Multi-qubit observables are computed from bitstrings
- Avoids additional unitary transformations
- Reduces noise accumulation

Thus, full-register measurement is optimal under realistic hardware constraints. □

Key question

Problem

Can we reconstruct expectation values of a Hamiltonian using only single-qubit measurements?

We show:

- This fails in general
- It fails even for simple two-qubit systems

Theorem: Failure of marginal reconstruction

Theorem

There exist quantum states ρ and observables H such that the expectation value $\langle H\rangle$ cannot be reconstructed from all single-qubit measurement statistics.

Proof.

Consider a two-qubit system and the observable

$$H = Z_1 Z_2.$$

Take two states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad \rho_{\text{mix}} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|).$$

Both have identical single-qubit marginals:

$$\langle Z_1\rangle = \langle Z_2\rangle = 0.$$

Explicit VQE counterexample

Consider Hamiltonian:

$$H = Z_1 Z_2.$$

Ansatz:

$$|\psi(\theta)\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle.$$

Exact energy:

$$E(\theta) = \langle Z_1 Z_2\rangle = 1.$$

Single-qubit estimate:

$$\langle Z_1\rangle = \cos^2\theta - \sin^2\theta, \quad \langle Z_2\rangle = \cos^2\theta - \sin^2\theta.$$

Naive reconstruction:

$$\langle Z_1 Z_2\rangle \approx \langle Z_1\rangle\langle Z_2\rangle = (\cos^2\theta - \sin^2\theta)^2.$$

Failure

This is **incorrect** except at special points.

Error in VQE energy landscape

True energy:

$$E_{\text{true}}(\theta) = 1.$$

Naive single-qubit estimate:

$$E_{\text{naive}}(\theta) = (\cos 2\theta)^2.$$

- True landscape: flat
- Naive landscape: oscillatory

Consequence

Optimization will converge to **wrong minima**.

Theorem: VQE failure under marginal approximation

Theorem

Replacing multi-qubit correlators with products of single-qubit expectations leads to incorrect energy landscapes and can produce false minima in VQE.

Proof.

Let $H = Z_1 Z_2$.
True:

$$E(\theta) = \langle Z_1 Z_2 \rangle.$$

Approximation:

$$E'(\theta) = \langle Z_1 \rangle \langle Z_2 \rangle.$$

Since correlations are neglected:

$$E'(\theta) \neq E(\theta),$$

and the optimization problem changes.
Thus the minimizer of E' is not the minimizer of E. □

Numerical experiment

Goal

Demonstrate failure of marginal reconstruction numerically.

We compare:

- Exact expectation value
- Approximation using marginals

Python code

import numpy as np

def state(theta): return np.array([np.cos(theta), 0, 0, np.sin(theta)])

Z = np.array([[1,0],[0,-1]]) Z1Z2 = np.kron(Z, Z)

def expectation(state, operator): return np.real(state.conj().T @ operator @ state)

def single$_{q}ubit_{e}xpectations(state) : rho = np.outer(state, state.conj())Z1 = np.kron(Z, np.eye(2))Z2 = np.kron(np.eye(2), Z)return(np.trace(rho@Z1), np.trace(rho@Z2))$

thetas = np.linspace(0, np.pi/2, 50)

true$_{v}als = []approx_{v}als = []$

for theta in thetas: psi = state(theta)

exact true$_{v}als.append(expectation(psi, Z1Z2))$

marginals z1, z2 = single$_{q}ubit_{e}xpectations(psi)approx_{v}als.append(z1 * z2)$

print("True:", true$_{v}als[: 5])print("Approx : ", approx_{v}als[: 5])$

Result of numerical experiment

- Exact value:

$$\langle Z_1 Z_2 \rangle = 1$$

- Approximation:

$$(\cos 2\theta)^2$$

Conclusion

Marginals fail to reproduce correlations.

Connection to VQE measurement problem

Modern VQE requires:

$$H = \sum_k c_k P_k$$

Each term:

$$\langle P_k \rangle$$

is estimated from **joint measurement outcomes**.

Important fact

Grouping and simultaneous measurement reduce cost but still rely on multi-qubit outcomes $[\mathrm{oai}_c itation : 0arXiv](https : //arxiv.org/abs/1907.03358?utm_source = chatgpt.com)$

Final conclusion

- Single-qubit measurements give only marginals
- Correlations are essential for quantum advantage
- VQE fails if correlations are ignored
- Full-register measurement is fundamentally necessary

Key insight

Quantum algorithms operate on **correlations**, not individual qubits.

Summary

- Quantum information is inherently **non-local**
- Single-qubit measurement loses correlations
- Mapping observables increases circuit depth
- Modern hardware supports parallel measurement
- NISQ algorithms rely on full bitstring sampling

Key insight

Quantum computing is fundamentally about sampling high-dimensional distributions, not extracting single bits.